

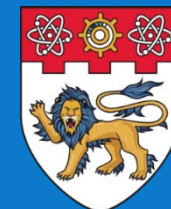


SC2002: Object-Oriented Design & Programming

Introduction (Second Half)

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Consultation Hours:

- Send an email for an appointment





Why OODP: Industry Examples and Applications

- **Enterprise Applications:** Major enterprise platforms (like Java Spring, .NET) leverage OODP principles to build large-scale, mission-critical systems.
- **Web and Mobile Applications:** Frameworks for both web and mobile development are often built on object-oriented concepts, making them more adaptable and easier to extend.
- **Cross-Domain Usage:** Whether it's in finance, healthcare, e-commerce, or telecommunications, companies rely on OODP to design systems that are robust, scalable, and easier to maintain.

Project development process:1

1. Requirements Gathering & Analysis

- **Stakeholder Engagement:**

Talk to users and stakeholders to understand needs.

- **Documentation & Prioritization:**

Write down requirements and focus on the most critical features.

Project development process:2

2. System Design & Architecture

- **Choose Architecture:**
Decide between styles like microservices or monolithic.
- **Modular Breakdown:**
Split the system into smaller, manageable parts.
- **Apply OODP:**
 - **Abstraction & Encapsulation:** Hide internal details.
 - **Inheritance & Polymorphism:** Promote code reuse and flexibility.
 - **SOLID Principles:** Keep code maintainable and scalable.
- **Integrate Design Patterns:**
Use patterns (e.g., Singleton, Factory, Observer) to solve common problems and improve modularity.

Project development process:3

3. Implementation

- **Coding Standards & Version Control:**
Follow best practices and use tools like Git.
- **Modular Development:**
Build self-contained classes and objects.
- **Testable Code:**
Design code that's easy to test and integrate.

Project development process:4

4. Testing & Quality Assurance

- **Testing:**
Run unit and integration tests to ensure each module works.
- **Continuous Integration (CI):**
Regularly merge code to catch issues early.
- **Design for Testability:**
Structure code to simplify automated testing.

Project development process:5

5. Deployment & Maintenance

- **Continuous Deployment (CD):**
Automate the release process.
- **Monitoring & Feedback:**
Use tools to track performance and gather user feedback.
- **Iterative Improvements:**
Regularly update and refactor the system using OODP and design patterns.

Today (2020s reality)

Across industry reports, practitioner surveys, and engineering experience:

- **Maintenance + evolution** typically consumes:
 - **40–60%** in well-run teams
 - **60–80%** in long-lived, complex systems (finance, healthcare, games, infra)

“But we have AI now — shouldn’t this go down?”

This is the *key misconception*.

AI reduces certain costs, but does not remove maintenance.

Project development process:2

2. System Design & Architecture

- **Choose Architecture:**

Decide between styles like microservices or monolithic.

- **Modular Breakdown:**

Split the system into smaller, manageable parts.

- **UML Modeling:**

Class Diagram: Abstraction & Encapsulation

Create a class diagram to show the key classes and their relationships.

UML diagrams are essential for documenting the design decisions made using OODP principles and the specific design patterns, ensuring that the structure and interactions are well understood by the development team.

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- **Integrate Design Patterns:**

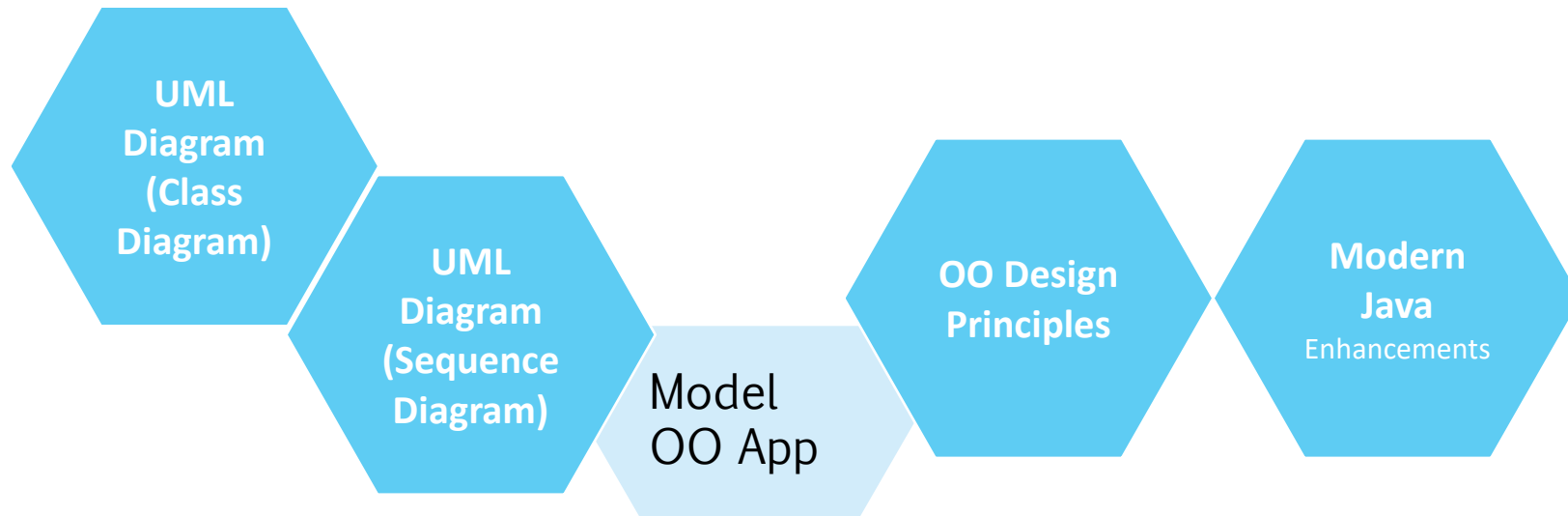
Sequence Diagram:

Use patterns (e.g., Singleton, Factory, Observer) to solve common problems and improve modularity. Develop sequence diagrams for key use cases. These diagrams detail how objects interact over time, showing the flow of messages and the dynamic behavior of the system. They help validate design decisions and ensure that interactions between components are well understood.

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Course Outline

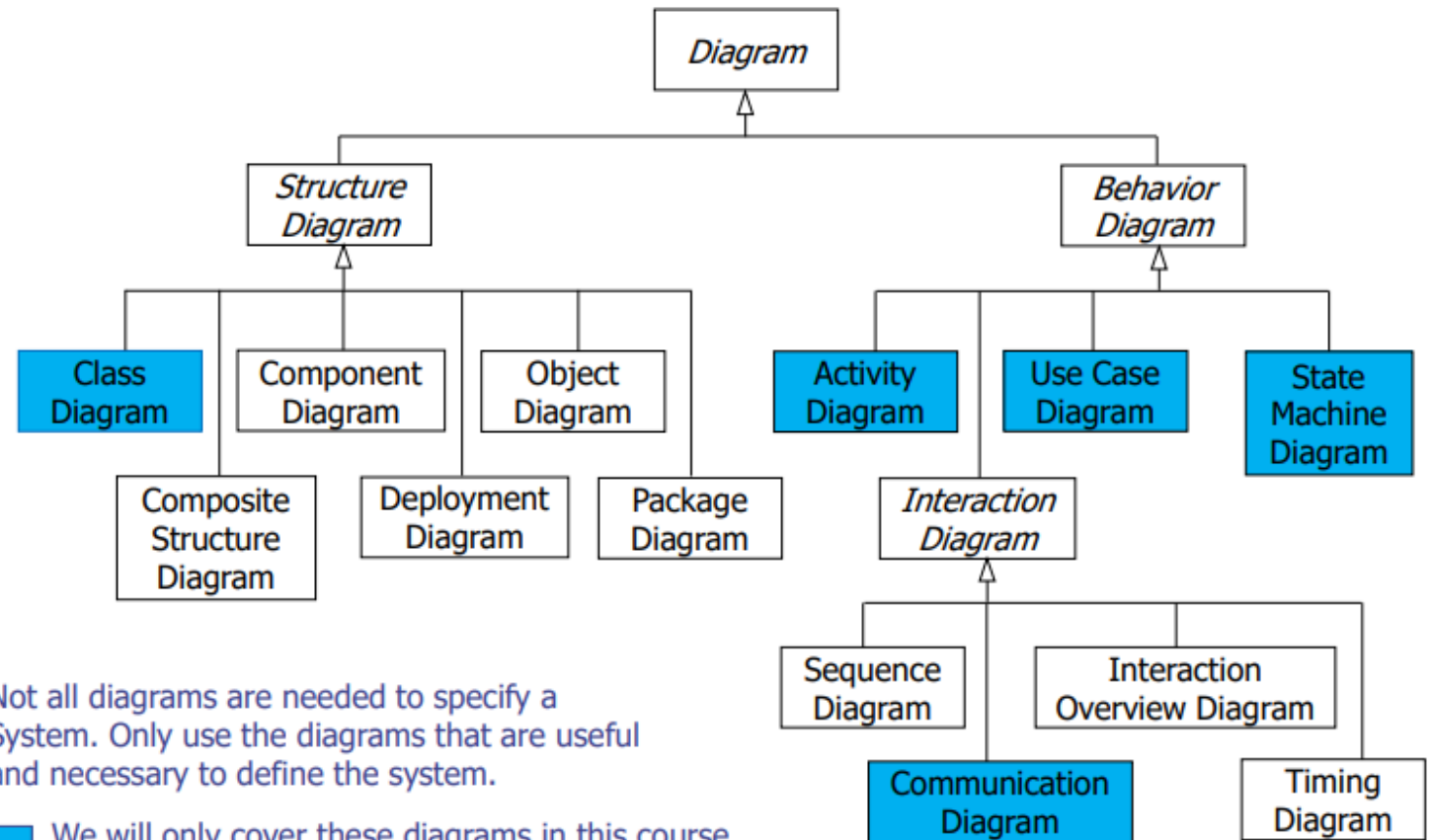
The following topics will be covered during the second half of the course:



UML Diagram Types

CZ 3002 – Advanced Software Engineering

Course Code	CZ3002
Course Title	Advanced Software Engineering
Pre-requisites	CZ/CE2006: Software Engineering



Not all diagrams are needed to specify a System. Only use the diagrams that are useful and necessary to define the system.

 We will only cover these diagrams in this course (Others will be covered in CZ3003)

In lecture, the following softwares will be used in demo.

- Eclipse for Java programming. (Same as first half)
- Visual Paradigm for UML diagram drawing (you are free to use your preferred software)
 - The course labs are all installed with the latest VP for your usage. You can also download @ <https://www.visual-paradigm.com/download/>
- **license key:** The license key for UML 17.2 Standard Edition is 75Q84-4B65D-Y6WDY-24XF8-ANSHF
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- **which is free for CCDS students since CCDS has purchased academic license for all CCDS students.**
 - You need download version 17.2 and use your NTU email to register for the verification code.
- You may find the following resources helpful if you are new to Visual Paradigm
 - How to Draw class diagram using visual paradigm || Step by step: <https://www.youtube.com/watch?v=Hl6Fa1QGxQo&t=3s>
 - How to Draw Class Diagram? https://www.visual-paradigm.com/support/documents/vpuserguide/94/2576/7190_drawingclass.html
 - Official user guide @ <https://www.visual-paradigm.com/support/documents/vpuserguide.jsp>



VP UML L

UML
Diagram
(Class
Diagram)

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	TEL Video under Content folder	Face-to-face sessions (Tuesday)	Online Q&A Session (Thursday)	Tutorials
Week 8	Introduction: Second Half Week 8: UML Model Class Relationships - Class Diagram	Weeks 8–12: One-hour face-to-face revision lecture per week (Tuesdays, 11:30 am–12:30 pm). The sessions will include MCQ practice, explanations of key and challenging concepts, targeted revision questions to apply and recap learned material, and final revision.	Weeks 8–12: One-hour Q&A session per week (Thursdays, 11:30 am–12:30 pm). The sessions will be conducted online via Teams , following the same arrangement as the first half of the semester, and will focus on clarifying students' doubts through Q&A.	Tutorial 5: Inheritance & Polymorphism
Week 9	Week 9: Sequence Diagram & ModelOOApp			Tutorial 6: Class Diagram
Week 10	Week 10: Design Principles			Tutorial 7: Sequence Diagram
Week 11	Week 11: Modern Java Enhancements			Tutorial 8: Design Principles (e-learning)
Week 12	Week 12: -Review of students' questions -quiz revision			Tutorial 9: Modern Java Enhancements
Week 13,14	NO classes. Assignment work. Assignment Consultation, if necessary. Week 13 Sunday Group Assignment code and doc submission deadline Week14 Thursday morning quiz2			

Course Assessment Components

5% Online quiz on Basic Java Syntax

45% Two Physical Quizzes

- One in Week 7 (Done)
- One in Week 14 (Thursday April 23)

10% Lab completion

35% Group assignment

5% Tutorial attendance

Quiz 2: Online MCQ quiz

- **Date: Week 14 (Thursday April 23)**
- **Time: 9:30-10:45am.**
- **Duration: 1.25 hours (75 Mins).**
- **Question type: MCQ. Single (default)/Multiple correct (state specially) answers MCQs.**
- **Content: All lecture content.**
- **Number of questions: 37.**
- **Venue: CCDS labs.**
- **Closed book. Lockdown browser is required.**

Group Assignment Assessment components

- UML Class Diagram
- UML Sequence Diagram
- Design Consideration
- Implementation Code
- Report

